higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

15 August 2024 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and 1 formula sheet.

303Q1G2415

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A: FLOW MEASUREMENT

QUESTION 1

1.1 Name THREE types of Pitot tubes. (3)

1.2 Name FOUR precautionary measures to be considered when using a Pitot tube (4)

1.3 Define each of the following terms:

1.3.1 Turbulent flow (2)

1.3.2 Laminar flow (2)

1.4 The Venturi tube is a highly accurate primary measuring device used to create a difference in pressure in flowing fluid.

1.4.1 Make a neat, labelled sketch of a Venturi tube. (5)

1.4.2 What is the main factor limiting the size of the instrument in QUESTION 1.4.1? (1)

[17]

QUESTION 2

2.1 State FOUR precautions to observe during the installation of weirs. (4)

2.2 Give ONE word or term for each of the following descriptions. Write only the answer next to the question number (2.2.1–2.2.6) in the ANSWER BOOK

2.2.1 The device that controls the amount of flow of a substance.

2.2.2 The position where the pressure reaches a minimum value after the liquid has passed through the restriction.

2.2.3 An obstruction placed in an open channel at a right angle to the direction of flow to create a pressure head.

2.2.4 An instrument that measures the total volume of flow in a certain time.

2.2.5 A velocity-sensitive device that measures the volumetric flow of gases and liquids.

2.2.6 The device that creates a difference in pressure in a flowing fluid.

(6 × 1) (6)

2.3 Name TWO types of weirs. (2)

[12]

TOTAL SECTION A: 29

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SECTION B

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY MEASUREMENT

3.1 Name FOUR types of buoyancy systems where a float or displacer is immersed in the liquid and the measuring system is influenced by the changes in density of the fluid. (4)

3.2 Make a neat, labelled sketch of a cone-and-plate viscometer. (5)

3.3 Study FIGURE 1 and answer the questions that follow.

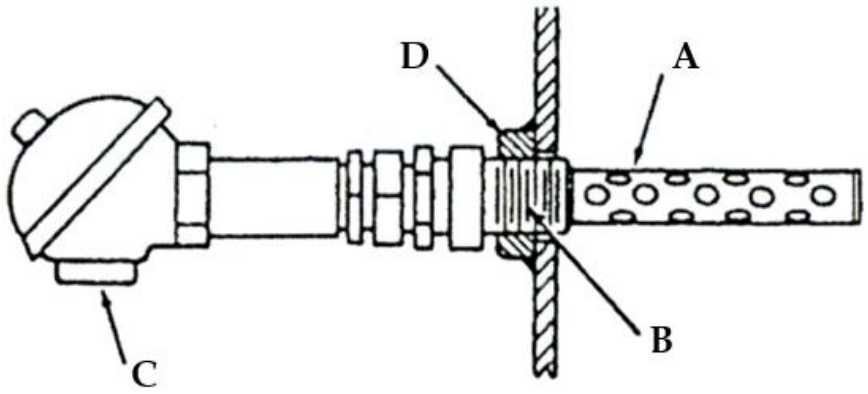


FIGURE 1

3.3.1 What type of hygrometer is represented in FIGURE 1? (2)

3.3.2 Name the labelled components in FIGURE 1 by writing only the answer next to the letter (A–D) in the ANSWER BOOK. (4)

3.4 Name THREE points that need to be observed when using a cone-and-plate viscometer . (3)

3.5 Study the sketch in FIGURE 2 and answer the questions that follow.

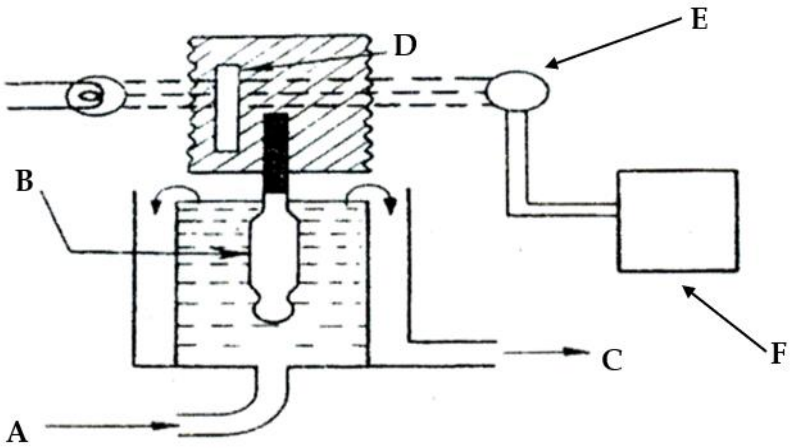


FIGURE 2

3.5.1 What type of hydrometer is represented in FIGURE 2? (2)

3.5.2 Name the labelled components in FIGURE 2 by writing only the answer next to the letter (A–F) in the ANSWER BOOK. (6)

3.6 Name SIX undesirable things to watch out for when using coaxial-cylinder rotational viscometers. (6)
[32]

QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 The material deposits on the glass electrode membrane inhibits the electrode response and sensitivity.

Mention FOUR ways of reducing the deposition rate on the glass electrode membrane. (4)

4.2 Numerous methods can be used to keep electrodes clean, such as solvent cleaning, because it is necessary for all types of deposit.

4.2.1 Name another technique that can provide continuous cleaning without introducing solvents**.** (1)

4.2.2 Explain how this method cleans electrodes. (3)

4.3 Several circuits are used for measuring electrical conductivity, with the AC Wheatstone bridge being the most widely applied and potentially the most stable and accurate.

Make a neat, labelled sketch of this circuit showing the voltage source, null balance indictor and the rebalancing mechanism. (5)

4.4 Solution conductivity varies with concentration and temperature in the AC Wheatstone bridge.

What is the function of the temperature compensator? (3)
[16]

QUESTION 5: AUTOMATIC CONTROL

5.1 To control a process, one must able to discriminate between the set point and the measured variable.

Name TWO methods of discrimination commonly used in pneumatic controllers. (2)

5.2 Refer to FIGURE 3 and answer the questions that follow.

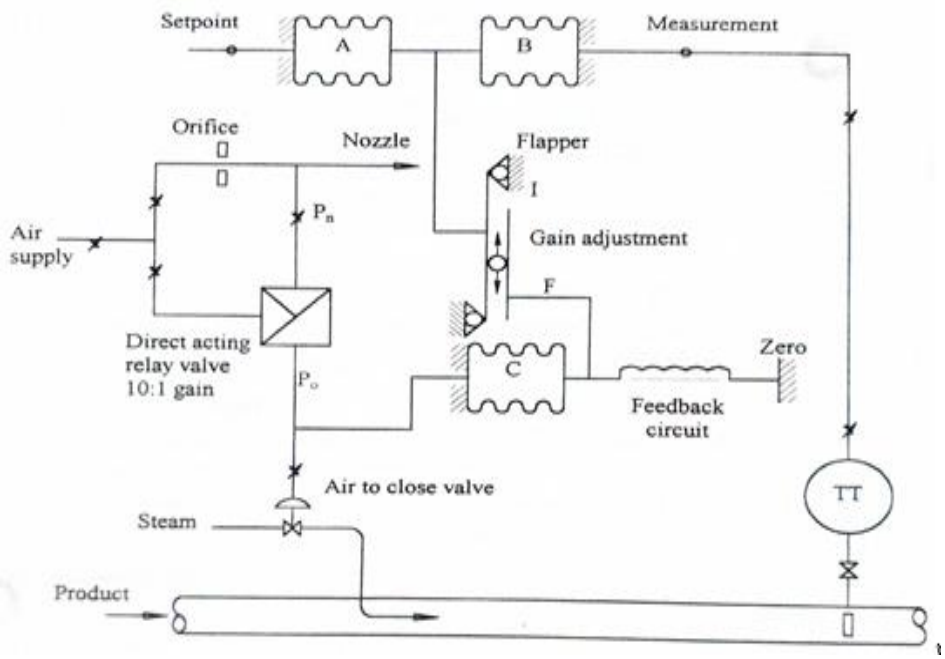


FIGURE 3

5.2.1 Name the type of control action shown in FIGURE 3. (2)

5.2.2 What is the function of bellows A and B? (2)

5.2.3 Define the *controller action* depicted in FIGURE 3. (2)

5.2.4 What is the main disadvantage of this type of controller? (2)

5.2.5 What action can be taken to overcome the disadvantage of the controller shown in FIGURE 3? (3)

5.2.6 When an error occurs, the recovery time for the system to stabilise is long.

What can be done to minimise this lag? (2)

5.2.7 How will this controller react when the measured variable increases more than the set point? (8)

[23]

TOTAL SECTION B: 71
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$$Q = \frac{8}{15} \quad Tan \frac{\theta}{2} \sqrt{2g.H^5}$$

$$Q = \frac{2}{3} \; B\,C\sqrt{2g.H^3}$$

$$Q = \frac{2}{3} \; CB\sqrt{2g.H^3}$$

pH = -log[H^+]

pOH = -log [OH^-]

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